# Week 14

**Course Level Objectives:**

- Perform hypothesis testing using statistical methods. (CLO 8)

| Module Objectives | Instructional Materials | Learning Activities | Assessments | Tools |
|---|---|---|---|---|
| **1.** State the hypotheses for a chi-square test of association from a two-way table. (CLO 8)<br>**2.** Compute the expected counts for a two-way table. (CLO 8)<br>**3.** Check that all expected counts are at least 5. (CLO 8)<br>**4.** Compute the chi-square statistic and its degrees of freedom, $(r-1)(c-1)$. (CLO 8)<br>**5.** Interpret the right-tail p-value of a chi-square test. (CLO 8)<br>**6.** Relate the chi-square test on a 2×2 table to the two-proportion z test. (CLO 8)<br>**7.** Explain why a significant result shows association but not direction or causation. (CLO 8) | e-Text: 7.2 Testing for an Association Between Two Categorical Variables<br>Interactive Lecture: 7.2 Testing for an Association Between Two Categorical Variables<br>Slides: 7.2 Testing for an Association Between Two Categorical Variables<br>StatKey – Theoretical Distributions ($\chi^2$) | Participation: Chi-Square Test – Your Turn: Faculty Rank and Gender: compute expected counts, cell contributions, $\chi^2$ and df; decide and interpret | Online Homework: 7.2 Testing for an Association Between Two Categorical Variables (due Sun Nov 29, 11:59 PM) | WileyPLUS<br>StatKey<br>Blackboard |

**Also this week:** Test 3 Retake – Mon Nov 23 – Sun Nov 29 (optional; higher score counts)